

Murtaza S. Ahmed

Embedded & Distributed Systems Engineer (Backend focus)

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Computer Engineering student specializing in embedded edge AI systems, distributed systems development, and backend infrastructure. Experienced in TensorRT-accelerated inference pipelines on NVIDIA Jetson Nano, decentralized media provenance protocols using IPFS/Arweave, and deterministic Linux system provisioning.

Professional Experience

Undergraduate Research Assistant (Edge AI Systems)

Dec 2025 -- Present

Department of Mechanical Engineering (MJCET)

Hyderabad, TG

- Developed an embedded edge AI platform using an NVIDIA Jetson Nano, IMX719 camera module, and telemetry sensors.
- Optimized computer vision inference pipelines using TensorRT FP16 acceleration, achieving 40-70 ms latency (14-20 FPS) under constrained compute environments.
- Implemented deterministic execution and monitoring systems for stable long-duration deployment.
- Researched adaptive workload scheduling and hardware-aware resource allocation strategies for heterogeneous LLM inference systems in interdepartmental R&D program.
- Exploring fault-tolerant orchestration approaches for distributed AI workloads across mixed compute environments.

Builder & Systems Architect

July 2025 -- Present

Verity (Decentralized Verification Platform)

Hyderabad, TG

- Designing and implementing system to verify digital media provenance using blockchain and decentralized storage.
- Integrating AI modules for manipulation detection and automated claim extraction.
- Developing media provenance infrastructure using distributed storage systems including IPFS and Arweave.
- Building concurrent ingestion pipelines and immutable audit logging mechanisms for ML-based verification workflows.

Backend Infrastructure Engineer

Sept 2024 -- Present

Freelance Consulting

Hyderabad, TG

- Migrated client applications to containerized Linux deployment environments with automated CI/CD workflows.
- Designed backend APIs and optimized PostgreSQL schemas for improved concurrency and deployment reliability.

Projects

Smart Agri Four-Legged Bot (SAFL-B) (C, Python, UART/Serial, I2C, GPIO, Hardware Control)

Dec 2025

- Awarded 1st Place at the MAKEFORHYDERABAD 2-Day Makeathon for autonomous robotics systems integration.
- Implemented deterministic UART/Serial communication protocols for multi-joint motor control.
- Developed telemetry stabilization routines to reduce terrain-induced sensor and image noise.

Deterministic Workspace Provisioning (Bash, Linux, Arch Linux, Podman, Git)

June 2025

- Built automated provisioning pipelines for lightweight Linux workstation deployment using Arch Linux and shell automation.
- Configured container runtimes (Docker/Podman) and developer tools to deploy reproducible workspace layers.

Technical Skills

Languages:

C, Python, Bash, PostgreSQL, Git

Systems:

Linux, Arch Linux, Docker, Kubernetes, Podman

Hardware & Interfaces:

UART/Serial, I2C, GPIO

Education

- Relevant Coursework: Scientific Computing, Embedded Systems Programming, System Modeling & Simulation